# StrataWave Uyum Durumu ve DEEPSOIL Example Parity Raporu

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Bu rapor, mevcut repo uzerindeki native StrataWave run'lari ile DEEPSOIL 7 batch referans ciktilarinin dogrudan karsilastirilmasini ozetler. Ama amac tam DEEPSOIL esdegerligi iddia etmek degil; mevcut parity seviyesini dürüstce gostermektir.

## Executive verdict

StrataWave ana analiz yollari ve OpenSees effective-stress adapter'i calisiyor. DEEPSOIL parity ise su an kismi seviyede. En iyi mevcut vaka, nonlinear_5a_rigid_dt0025_tuned icin PSA NRMSE=0.1939 veriyor; buna ragmen time-history korelasyonu zayif kaldigi icin bu sonuc yalnizca PSA-seviyesinde kisimli uyum olarak yorumlanmalidir.

| Bilesen | Durum | Not |
|---|---|---|
| Native linear/EQL/nonlinear | Calisiyor | Example pack ve native run artifact'leri var. |
| OpenSees effective-stress adapter | Calisiyor | Smoke/reference run artifact'leri mevcut. |
| DEEPSOIL parity | Kismi | En iyi vaka nonlinear_5a_rigid_dt0025_tuned ve bu vaka PSA seviyesinde kisim |
| Native full effective-stress solver | Eksik | OpenSees adaptoru var, native u-p solver yok. |

## Case matrix

| Case | Boundary | dt_sw | Surf NRMSE | Surf Corr | PGA diff % | PSA NRMSE | Verdict |
|---|---|---|---|---|---|---|---|
| linear_1b | rigid | 0.0200 | 0.4572 | -0.0120 | -23.1 | 0.2346 | partial |
| linear_1b_mkz0 | rigid | 0.0200 | 0.5858 | 0.0004 | -12.3 | 0.3216 | partial |
| nonlinear_5a | elastic_halfspace | 0.0200 | 5.0995 | 0.0131 | 524.0 | 3.7486 | poor |
| nonlinear_5a_rigid | rigid | 0.0200 | 5.0937 | 0.0131 | 523.3 | 3.7470 | poor |
| nonlinear_5a_rigid_dt005 | rigid | 0.0050 | 0.4426 | 0.0018 | -57.4 | 0.2026 | partial |
| nonlinear_5a_rigid_dt0025 | rigid | 0.0025 | 0.4389 | -0.0032 | -57.1 | 0.2007 | partial |
| nonlinear_5a_rigid_dt0025_tuned | rigid | 0.0025 | 0.4364 | -0.0015 | -68.4 | 0.1939 | partial |
| nonlinear_5a_elastic_dt005 | elastic_halfspace | 0.0050 | 0.4425 | -0.0005 | -63.9 | 0.2092 | partial |
| nonlinear_5a_elastic_dt0025 | elastic_halfspace | 0.0025 | 0.4384 | 0.0004 | -64.0 | 0.2074 | partial |

## Current findings

Best current case: Example 5A nonlinear rigid-base dt=0.0025 tuned (nonlinear_5a_rigid_dt0025_tuned). PSA NRMSE=0.1939, PGA diff=-68.40%, surface correlation=-0.0015.

Bu vaka, diger nonlinear ve linear denemelere gore daha makul bir spektral uyum gosteriyor. Ancak time-history tarafinda guvenli bir parity iddiasi icin yeterli degil.

### Technical caveats

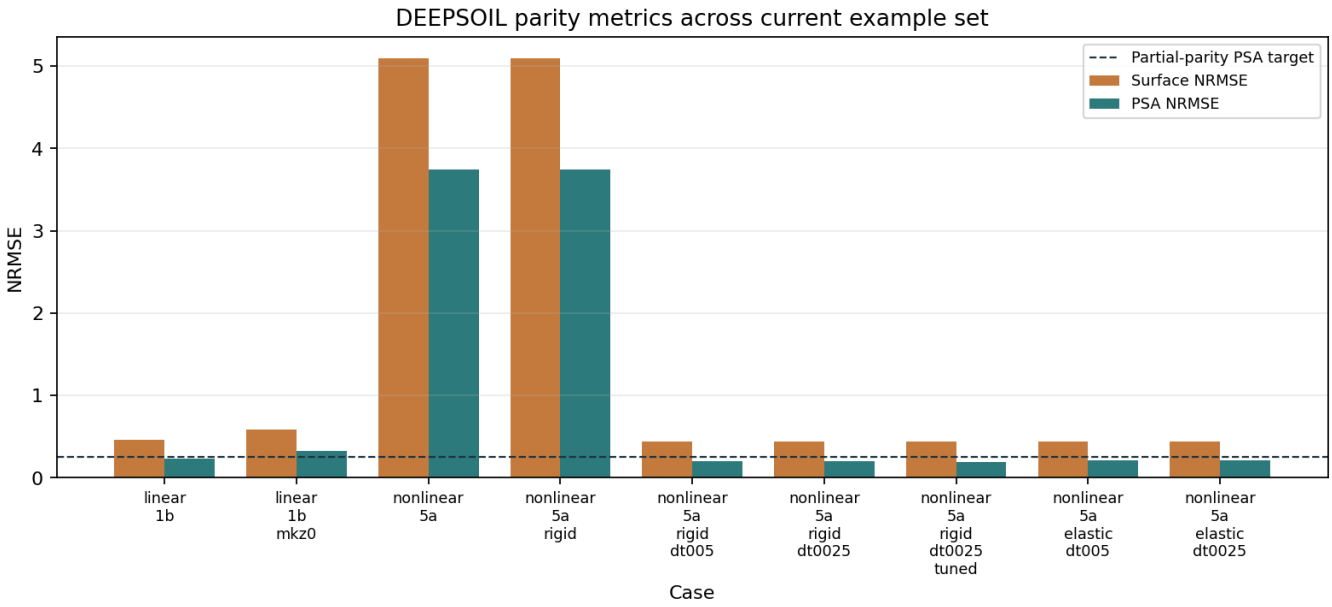
1) Native linear/nonlinear solver su an base-fixed odakli, oysa DEEPSOIL orneklerinin bir kismi elastic-halfspace kullaniyor.

2) Small-strain damping formulasyonu ve complex-modulus davranisi henuz birebir kapatilmadi.

3) Native MKZ/GQH patikasi calisiyor, fakat published-reference seviyesinde tam konstitutif parity kapanmadi.

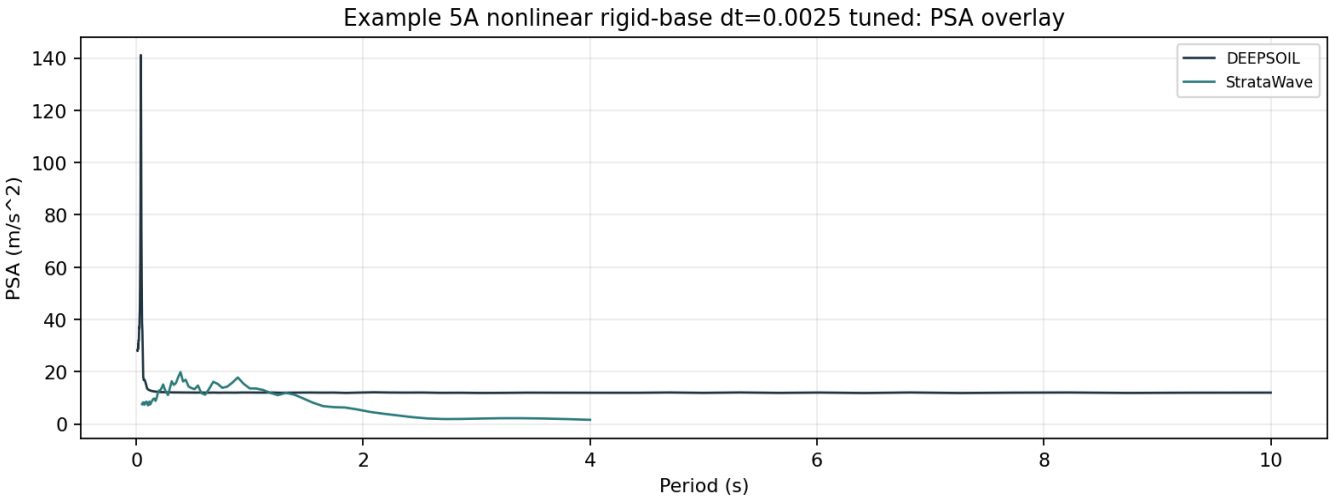
## Metrics overview

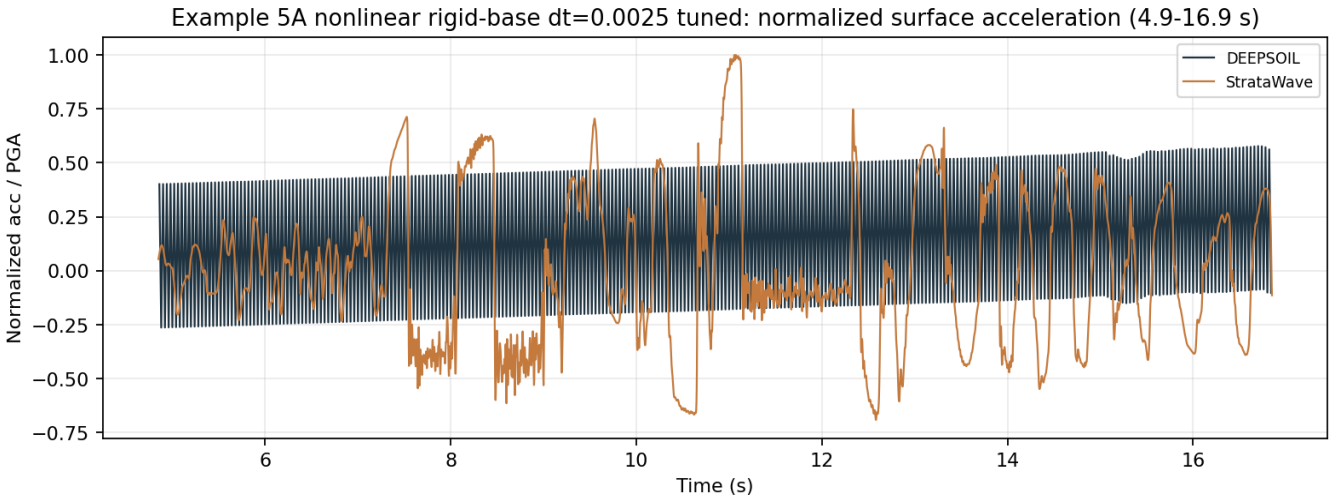
Asagidaki grafik, mevcut ornek setinde surface ve PSA NRMSE dagilimini gosterir. NRMSE dusuk oldukca parity iyilesir. Raporun mevcut yorumu icin PSA NRMSE daha agirlikli kullanilmistir.




## Best-case overlays

PSA overlay, en iyi mevcut vakanin DEEPSOIL ile spektral benzerligini; surface acceleration overlay ise time-domain tarafindaki zayifligi ayni anda gosterir.






## Conclusion

Mevcut repo, hibrit hedefe uyumludur: native linear/EQL/nonlinear yollar calisiyor, OpenSees effective-stress adapter'i calisiyor, DEEPSOIL parity tooling mevcut. Ancak bu nokta tam DEEPSOIL esdegerligi degildir.

En iyi mevcut parity vakasi nonlinear_5a_rigid_dt0025_tuned olup, bu vaka icin PSA NRMSE=0.1939 ve PGA farki -68.40% seviyesindedir. Dolayisiyla bugunku dogru teknik ifade: DEEPSOIL-benzeri is akisi + OpenSees destekli effective-stress dogrulamasi + native MKZ/GQH cekirdegi olan hibrit analiz platformu.

### Best-case warnings

- Time-step mismatch detected (StrataWave=0.0025s, DEEPSOIL=0.02s).